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Predictive quality performance control in BPM: proposing a framework for predicting quality anomalies

Naef Saab^{a,b,*}, Remko Helms^a, Martijn Zoet^b

^aOpen University, Heerlen, The Netherlands

^bZuyd University of Applied Science, Heerlen, The Netherlands

Abstract

Business process management (BPM) literature suggests that more than 60% of quality improvement projects fail due to factors associated with the lack of predictive quality performance control and the failure of continuously searching for quality anomalies in quality performance over time. Quality anomalies are indications of extreme performance deviation from quality expectations and requirements. The findings suggest that quality performance control in BPM is the scientific method for producing quality anomaly knowledge and signalling opportunities for informed, systematic, and continuous performance improvement. A predictive framework is proposed based on the findings.

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Keywords: BPM; TQM; BPR; QFD; SPC; quality performance measurement; time series; temporal data mining; predictive analytics; outlier and deviation mining; process mining; quality control; quality improvement; design science research; business process intelligence

* Corresponding author.

E-mail address: naef.saab@zuyd.nl

1. Introduction

More than 60% of quality improvement projects in Business Process Management (BPM) fail due to factors associated with predictive quality control. Popular reasons for these failures include the inability to detect and predict quality performance problems continuously over time, the lack of long-term temporal performance focus from a stakeholder's perspective, the lack of stakeholder involvement in performance measurement, unstructured performance measurement systems (PMS), and difficulties with prioritizing attention towards solving quality performance problems continuously and proactively over time [1, 2]. Research has attempted to investigate quality problems in quality performance control and found gaps related to predictive anomaly detection in performance measurement as well, such as the inability to predict temporal anomalies which are indexed over time; the focus on point anomalies only; and the limited use of available tools, methods and techniques to detect anomalies [3-5]. From a methodological perspective, Choong [6] notes that organizations need a continuous process for quality performance control in BPM for detecting performance anomalies or problems, which defines stakeholders and quality standards, monitors and controls progress against quality standards, and improves processes towards and beyond quality standards sustainably and continuously over time [7]. From a quality improvement perspective, quality control in BPM addresses at least five dimensions of quality performance problems: time, compliance, nature of performance measurement, efficiency and effectiveness performance expectations, process design, and product design [2]. This research therefore addresses the following research question and associated research objectives:

Which quality performance control framework could be proposed for predicting quality performance anomalies in BPM over time?

1. Understand the role of quality control and quality improvement in BPM;
2. Understand the analytical methods for implementing quality control and quality improvement;
3. Understand analytical methods for detecting and predicting anomalies in data indexed by time units.

2. Quality Anomalies and Quality Performance Control in BPM

Studies have shown that quality performance of BPM capabilities has both positive and negative effects on financial performance and stakeholder satisfaction [8, 9]. BPM capability is the ability to integrate resources, knowledge, technology, processes, strategy, and other capabilities in alignment with stakeholders' objectives, expectations, requirements, and satisfaction levels [10]. In line with this definition, this research defines quality performance as the degree of fulfilling critical-to-quality performance requirements and expectations (CTQs), which satisfy stakeholders [11]. Stakeholders include but are not limited to customers, management, process owners, product owners, and suppliers [12, 13]. Their roles could be classified as collaborators, recipients, influencers, and claimants [13]. From a stakeholder's perspective, Golder, Mitra and Moorman [9] propose furthermore that quality performance is a set of three different states consisting of perceived quality, produced quality, and evaluated quality. Their proposed framework suggests that quality performance is perceived from a stakeholder's perspective, is a process with output, and is subjected to performance control, evaluation, assessment, diagnostic, monitoring, or auditing [9, 13]. Therefore, quality performance is multidimensional, subjective, and interpreted in a social context, consisting for example of continua between socially accepted value (good) and waste (bad) and between processes and outcomes [14]. These dimensions are in line with ethical, economical, legal, and philanthropic roles associated with stakeholder, agency and social responsibility theories [13]. Along these dimensions, extremely wide gaps between stakeholder CTQs and actual performance reflect high variations in quality performance and indicate quality performance problems, quality weaknesses, quality performance gaps, quality deficiencies, or quality anomalies [15]. Quality anomalies are detected with evaluation, diagnosis, assessment, auditing, monitoring, or controlling of the deviations of actual performance from CTQs, which are key activities of quality performance control [16].

From a performance measurement and empirical perspective in quality control, quality anomalies are quality performance problems. They are indicated by extreme deviations in performance metrics from CTQs [9]. Quality

anomalies indicate whether CTQs are extremely far from being met (i.e. negative quality anomalies) or are being extremely exceeded (i.e. positive quality anomalies) [2]. From a decision perspective, positive quality anomalies in quality performance are problems which suggest opportunities for capitalizing on excess value or adjusting CTQs upwards. Negative quality anomalies suggest opportunities for reducing excess waste or adjusting CTQs downwards. From an empirical perspective, quality anomalies could be measured with data as well [17]. Literature suggests three types of quality anomalies which stakeholders should detect in quality performance control over time. As pointed out by Ahmed, Mahmood and Islam [17], point anomalies refer to an individual instance in the data which exhibits extreme performance deviations from CTQs metrics. Collective anomalies is when multiple point anomalies in the data reflect extreme performance deviation from CTQs metrics simultaneously, such as multiple Enterprise Resource Planning (ERP) process instances experiencing quality anomalies simultaneously [11, 17-19]. Context anomaly is when data instances reflect extreme performance deviations from CTQs metrics in particular circumstances such as in particular seasons, months, years, days, experiments, improvements or implemented decisions [17, 20]. Point anomalies have received the most attention in BPM literature [21].

As the occurrence of quality performance anomalies transcends functional boundaries of business organizations, all management disciplines are faced with the quality performance control challenge of detecting and managing quality anomalies. BPM is one particular example which encounters quality performance anomalies. BPM has received wide attention as a holistic and cross-functional discipline in business management with a focus on technology, processes, stakeholders, quality and sustainable performance [22, 23]. Three streams of research have defined the BPM discipline from a quality performance control and improvement perspective. Total Quality Management (TQM) takes a scientific, evidenced-based, and statistical approach for detecting and reducing quality anomalies through incremental quality performance improvement. Business Process Re-engineering (BPR) focusses on systematically reducing process anomalies through visualization and radical quality performance improvement [11]. Automation focuses on systematically reducing system anomalies through continuous quality performance improvement using information technology such as ERP systems, process mining, machine learning, business rules, workflow management, information systems, document management, and business intelligence [19, 24]. Based on these streams of research, BPM has become a capability where people, governance, processes, strategy, organization, information technology and performance management are continuously combined, orchestrated, monitored for quality performance anomalies, and improved in alignment with stakeholders and their CTQs [10].

3. Research Methodology

Based on the works of Dresch, Lacerda and Antunes Jr [25], Wieringa [26], and Sidorova and Isik [23], this study conducts a systematic literature review for proposing a framework. This review consists of thematic analyses for selecting relevant articles and theory grounding for addressing the research objectives and proposing an actionable predictive quality control framework for a visual evaluation. For each research objective, four quality criteria are employed.

Quality criterion 1: Articles published between the years 1996 and 2017 in peer-reviewed English language journals are retrieved through EBSCOHost API integration for university libraries, using Boolean search terms derived from the research question and objectives. EBSCOHost API accesses a variety of popular scientific databases such as IEEE Xplore Digital Library, ScienceDirect, SpringerLink, Wiley Online Library, Academic Search, JSTOR, Sage Premier, and Google Scholar. Mainly articles published in journals about computer science, performance, process, quality, business, analytics, statistics, BPM, TQM, and intelligence are considered.

Quality criterion 2: Following the examples of Sidorova and Isik [23] for thematic analyses, this study applies Latent Semantic Analysis (LSA) on the abstracts of the articles retrieved in quality criterion 1. LSA in this regard analyzes words in abstracts and generates latent themes using principal component analysis (PCA) in Python computer programming language. Latent themes are subsequently selected using elbows in the scree plots. K-means cluster analysis forms groups of article, based on the Euclidian distance. The number of groups are selected based on the maximum silhouette coefficients [27]. A word cloud per cluster subsequently displays the most commonly occurring themes in the abstracts of the articles. Only the clusters of articles displaying high frequencies of the

themes related to quality, business, process, control, improvement, management, BPM, performance, framework, system, information, TQM, data, and modelling are selected for further analysis.

Quality criterion 3: The titles of the articles which are excluded in quality criterion 2 are reviewed to ascertain if relevant topics were excluded. Subsequently, the selected articles are reviewed manually to ascertain their relevance in addressing the research question and objectives. Articles exhibiting generalizable research methods, conceptual frameworks, and systematic literature reviews are subsequently selected for further analysis.

Quality criterion 4: The frameworks, models, and tools of articles are grouped and compared for theory grounding, which consists of comparing methodologies and identifying underlying theory for generating a framework consisting of a structural component for providing context, and a procedural component for guiding implementation [25].

4. Results and Discussion

4.1. Quality Control and Quality Improvement in BPM

Several factors in the BPM literature are proposed to create quality anomalies in BPM and drive the need for incremental, radical and continuous quality performance improvement, such as changing business environments, stakeholders CTQs, competition, external customers, laws and regulation, management systems, fads and fashion, and importance of quality management [28]. These changes could increase quality performance gaps and create quality anomalies, often continuously [29]. BPM stakeholders encounter continuous weakness in quality performance, which could decrease stakeholder satisfaction and motivate stakeholders to take action for improving quality performance continuously [30]. Quality control is an essential process in quality improvement. In their literature review about continuous improvement (CI), Sanchez and Blanco [28] highlights CI as a key feature of quality management. They furthermore suggest that CI is a cycle consisting of activities which should be repeated over time for maintaining stakeholder satisfaction levels. Research on CI are classified into nine categories such as concept definition, implementation of CI, drivers/obstacles, methodologies, culture, control, management philosophies, technology, innovation, and human resources [28]. In this regard, the BPM literature highlights the importance of CI methods and control activities from different stakeholder perspectives, which explains the variety of CI methods organized under the BPM umbrella [19]. Siha and Saad [31] for example reviewed the critical success and failure factors of four popular quality improvement methods used in BPM: Six Sigma, Benchmarking, Business Process Reengineering, and Process mapping. Morais, Kazan, Pádua et al. [22] compare seven BPM improvement cycles against a professional BPM framework and found little alignment with strategy and stakeholders. These methods however do aim to reduce differences between CTQs and actual performance using applied research, evidence, data, statistics, modelling, observation, facts, reasoning and learning [32, 33]. Tort-Martorell, Grima and Marco [34] highlights this in their evidence-based management approach for reducing quality anomalies in business management, which relies on data for making informed decisions regarding performance improvement. Jeston and Nelis [19] and Association of Business Process Management Professionals (ABPMP) furthermore propose a BPM cycle for systematically integrating and improving BPM capabilities and aligning these with stakeholders' requirements and expectations [22]. Literature suggests that these CI cycles are variants of Deming's Plan-Do-Check-Act (PDCA) cycle for quality improvement as illustrated in Table 1 [30, 35].

Table 1 – Comparing Improvement Cycles with the Scientific Method.

Scientific Method	Deming	Lean	Six Sigma	Evidenced-based Management	BPM	Process Performance Measurement	ABPMP	Design Science Research
Dresch, Lacerda and Antunes Jr [25]	Knouse, Carson, Carson et al. [35]	Oakland [11]	Oakland [11]	Tort-Martorell, Grima and Marco [34]	Jeston and Nelis [19], Morais, Kazan, Pádua et al. [22]	Choong [6], Kueng [36]	Morais, Kazan, Pádua et al. [22]	Wieringa [26]
Inductive Reasoning	Plan	Define	Define	Decisions	Foundation, Enablement	Define Business Process Goals and Target Values	Planning, Analysis, Design, Modelling Implementation	Understand, Suggest
Inductive Reasoning	Do	Review	Measure	Questions, Data	Launch	Measure Process		Develop

Deductive Reasoning	Check	Investigate, Verify, Reinforce	Analyze, Control	Analysis	Understand, Sustainable Performance	Assess Process Performance	Monitoring, Controlling	Evaluate
Inductive Reasoning	Act	Execute	Improve	Information	Innovate, People, Development, Implement, Realize	Conduct Process Improvement	Refinement	Apply

The PDCA cycle is a scientific methodology for generating quality anomaly knowledge, solving quality performance problems, and improving quality performance [6, 36]. Scientific methodologies are preferred in enterprises because they enable scientific knowledge production and validation, and they are more reliable and rigorous for making informed quality improvement decisions [34]. The desire to measure quality performance and achieve reliable quality performance improvement have demanded a more systematic approach to quality control, based on the scientific method [32]. The scientific method is a systematic approach for producing knowledge about phenomena with the purpose of filling knowledge gaps or solving practical problems [25, 32, 33]. The scientific method consists of inductive reasoning where limited observations and experiments are used to build CTQs that are not always valid, to identify potential quality performance problems, or to identify potential solutions which are not always perfect. The scientific method also consists of deductive reasoning which relies on empirical evidence and data to validate CTQs, potential problems and solutions. Quality performance control also relies on these reasoning methods, and these methods are commonly observed in auditing, process mining, process control, quality assurance, and performance measurement [37-39]. As such, quality performance control in BPM is the systematic generation of knowledge about quality anomalies in quality performance with the objective of improving quality performance of BPM capabilities. During the first inductive reasoning phase, quality control systematically produces tentative CTQs knowledge in BPM using limited evidence from interviews, observations and experiments. During the deductive reasoning phase, quality control relies on evidence, data and performance measurement for producing quality anomaly knowledge and validating CTQs. During the second inductive phase, validated quality anomaly knowledge is used for communicating quality performance anomalies and for driving potential ideas for quality performance improvement in BPM [19, 32].

4.2. Analytical Methods for Quality Control and Quality Improvement

Several analytical tools, capabilities, and techniques are used for building quality anomaly knowledge and conducting quality improvement, such as Quality Function Deployment (QFD), Balanced Scorecard, process mining, statistical process control (SPC), brainstorming, interviews, questionnaires, cause-and-effect diagrams, flow charts, Pareto charts, scatter plots, stratification, machine learning, statistical modelling, run and control charts, and check and data collections sheets [11, 19, 40]. These methods are often combined and play an important role in the scientific method and quality performance measurement [32, 34]. They are essential for inductive reasoning in terms of generating CTQs knowledge about BPM performance capabilities; for deductive reasoning in terms of validating this knowledge with empirical evidence, data, and performance measurement; and for inductive reasoning in terms of suggesting areas and potential ideas for improvement in quality performance of business capabilities [32]. These tools are relevant for producing and applying quality anomaly knowledge in BPM. Veit, Camargo, Dresch, Lacerda and Kipper [41] proposes two types of scientific knowledge production (KP) in BPM in this regard. Mode 1 KP refers to producing knowledge for improving the scientific method from an academic perspective and Mode 2 KP refers to producing knowledge aimed at solving practical problems using the scientific method [41]. Mode 1 KP entails systematically building academic knowledge about process modelling and improvement, process management, BPM methodologies, relationship and integration process, and process implementation and automation [6]. Mode 2 KP entails systematically building practical and actionable knowledge about performance indicators and measures, continuous improvement, supply chain, outsourcing, change management and innovation [6]. The scientific method also defines the nature of Mode 2 KP as can be seen with process performance measurement, PDCA, Lean, process mining, Six Sigma, evidenced-based management and design science research methods [25, 34]. Mode 2 KP is therefore a popular applied scientific method for quality performance control in BPM in terms of generating quality anomaly knowledge scientifically with the purpose of improving quality performance of BPM capabilities. Recently, business intelligence (BI) has gained momentum as combining Mode 1 KP for improving

scientific methods and Mode 2 KP for applying these scientific methods for quality control and improvement [25, 42]. BI consists of interconnected tools, people, processes, and technology, which are orchestrated in scientific discipline and systematically converted into actionable quality control knowledge for quality improvement. BI supports quality control and uses performance measurement methods systematically to identify quality performance anomalies, drive quality performance improvement, and validate quality improvement. In this regard, business process intelligence (BPI) is increasingly considered as a capability and method for generating and applying quality anomaly knowledge systematically in quality control using performance measurement frameworks, information communication technology (ICT), process mining, statistical process control, and machine learning in BPM [42].

4.3. Predicting Quality Performance and Detecting Quality Anomalies over Time

One limitation of current quality performance control methods in BPI is the reactive focus on historical and temporal data which are indexed over time and the reactive focus on taking actions after quality anomalies have occurred [11]. Continuous monitoring has become popular in this regard; however, stakeholders are still inclined to improve quality performance reactively. Using time-series or temporal data, proactive quality control predicts quality performance in the future and detects quality anomalies continuously for timely quality performance improvement [37]. Time-series or temporal data are data points indexed by time units, such as hourly, daily, weekly, monthly, or yearly [43]. In order to predict the metrics underlying quality performance over time, the temporal data-mining literature suggests different techniques for predicting temporal data. Temporal data mining, which is often referred to as time-series analysis, is defined as the systematic production of knowledge from large amounts of time-series data using supervised, unsupervised, or reinforcement machine learning [44]. Liao, Chu and Hsiao [45] conducted a systematic review and summarized popular data-mining techniques such as support vector machine, a priori algorithm, artificial neural network, clustering, genetic algorithm, association rule, decision tree, neural network, classification, and feature selection. These techniques are useful for analyses which do not rely on statistical assumptions such as normal distribution, absence of autocorrelation, and linearity in the data. However, auto regressive integrated moving average (ARIMA) has been the most popular statistical technique for modelling linearity and autocorrelation in time-series data [46]. Autocorrelation or serial correlation is an issue when data points correlate over time. Temporal data are both linear and non-linear, are often correlated over time, and are not always normally distributed. Therefore, combining statistical models for modelling linear components with machine learning for modelling non-linear components is becoming a popular hybrid-modelling strategy in data science [47].

Machine learning and statistics play a key role in detecting quality anomalies in predictive quality control. Also known as outlier mining, deviation mining, novelty detection, or exception mining; anomaly detection techniques are helpful for identifying quality anomalies in quality performance in time-series or temporal data [48, 49]. Literature suggests several quality anomaly detection techniques for quality performance control. Agyemang, Barker and Alhaji [50] for example classify outlier mining techniques into numeric, cluster-based, and symbolic-based techniques. Numeric techniques are statistically based and are the most commonly used techniques used today in quality control and quality improvement, such as z-scores [11]. Cluster-based techniques rely on cluster analysis and finding small sized groups with extremely large or small values. Symbolic techniques convert unstructured data into numeric data for further analyses using numeric and cluster-based techniques [3, 17, 20, 51-54]. One common theme related to data mining from a machine learning and statistical perspective is the systematic process or scientific method which is deployed to produce knowledge from evidence and data using analytics. Known as the Cross Industry Standard Process for Data Mining, CRISP-DM consists of a method of understanding the business and its data, managing the data, developing the analysis model, evaluating the models for quality, and deploying the models, which is more in line with Mode 1 KP [44, 45, 55]. The Knowledge Discovery in Databases (KDD) model used for evidenced-based management extends the CRISP-DM cycle to practical applications related to Mode 2 KP. This is in line with design science research (DSR) for building academic knowledge and practical artifacts [42, 56].

5. Predictive Quality Performance Control Framework

This research proposes a predictive quality performance control framework with a structural component (1.0) for providing context, and a procedural component (2.0) for guiding implementation. As shown in Figure 1, the

structural component consists of three capabilities for supporting the procedure for predicting quality anomalies in quality performance with quality control. Firstly, the BPM capability (1.1) is essential and should consist of different dimensions such as people, strategy, performance management, processes, governance, organization, and ICT [10, 19]. Quality Function Deployment (QFD) is the second capability representing the quality improvement chain and consists of different Houses of Quality where CTQs are defined and translated to activities and quality performance control via a quality chain. The QFD capability (1.2) starts with defining the BPM stakeholders and their CTQs; converting these CTQs subsequently into products/services, processes, and activity design; and monitoring the performance of these components relative to the CTQs. The use of QFD methods and tools such as qualitative and quantitative research methods in BPM shifts the focus from performance management to quality performance management using scientific research. However, as the TQM literature suggests, the lack of technology and inability for continuous monitoring in quality performance management have also contributed to the failure of quality improvement projects [11]. The growing amount of data, technology, and analytical methods has provided more opportunities for information technology in BPM and TQM. Business Process Intelligence (BPI) in this regard is the third capability which combines different dimensions of the BPM capability and QFD for producing and applying Mode 1 and 2 process knowledge systematically. The BPI capability (1.3) is further characterized by understanding stakeholders' CTQs in BPM, understanding BPM data, managing BPM data, producing anomaly knowledge and artifacts with statistical modelling and machine learning, evaluating knowledge and artefacts, and applying produced knowledge and artifacts to quality performance problems in BPM [42]. The proposed framework assumes that BPM, QFD, and BPI are knowledge capabilities already in place and should be checked for before continuing with the procedural component. These structural components align with the contextual capabilities in Wieringa [26] DSR framework: organization (BPM), social (QFD) and knowledge contexts (BPM, QFD and BPI).

Design Science Research (DSR) is a systematic, applied, scientific and actionable research method for producing Mode 1 and 2 knowledge and artifacts systematically with the aim for solving practical problems with rigor and relevance using inductive and deductive reasoning [25]. Artifacts are the predicted quality anomalies. The procedural component in Figure 1 is reflected by a DSR cycle, which aligns with Deming's scientific PDCA cycle and Choong [6], Kueng [36] process performance measurement methods as seen previously in Table 1. On the Plan phase, practitioners start with determining BPM stakeholders, understanding their needs, and designing tentative CTQs about BPM capabilities using inductive reasoning such as interviews, case studies, surveys, focus groups, and literature review. On the Do Phase, CTQs are translated into performance target measures and operationalized into metrics in partnership with BPM, QFD, and BPI stakeholders. Performance metrics are subsequently operationalized into data requirements and data are retrieved with BPI stakeholders. Temporal or time-series performance data are captured, extracted and indexed by time units such as hourly, daily, weekly, monthly, and yearly. On the Study phase, quality performance predictions per time unit are generated and evaluated for future time units using statistical modelling [46] and machine learning [45] in BPI. Predictive quality performance is also determined on this phase as the difference between CTQs target metrics and predicted performance metrics, which are indexed by time units. Quality performance metrics are normalized with statistical z-scores and point quality anomalies are detected based on six sigma cut-off value [17, 20, 54]. Identifying groups of point anomalies in specific context provides collective and contextual anomalies. On the Act phase, quality performance anomalies are communicated with BPI to stakeholders and systems as DSR knowledge and artifacts output.

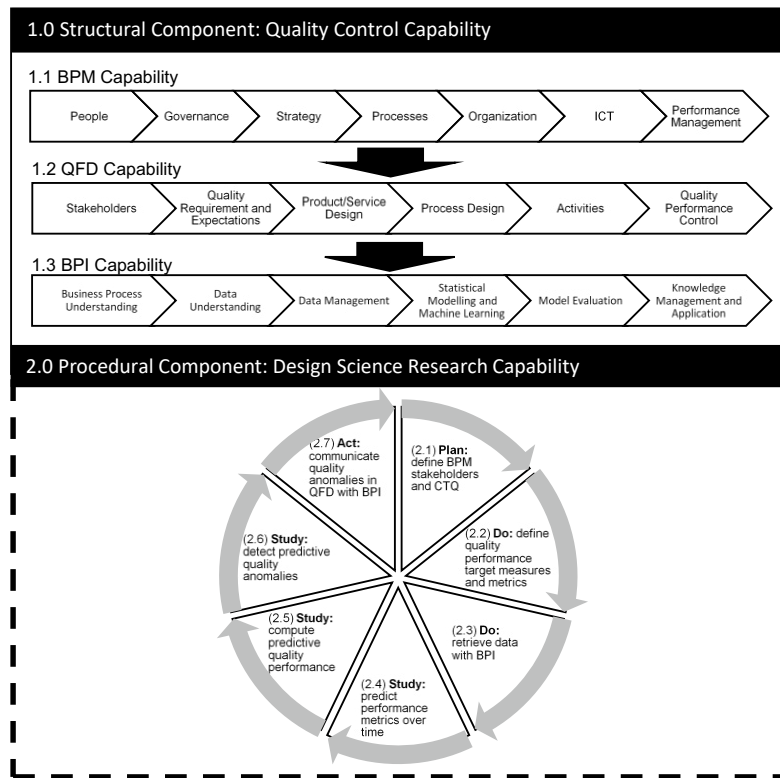


Fig. 1 - Predictive Quality Performance Control Framework for BPM

6. Limitations and Avenues For Future Research

As more companies begin to take a process approach to business management, future research should validate and improve the proposed framework in Figure 1. Surveys and focus groups with BPM professionals for example could generate data to validate and improve this framework. A case study could also be useful for testing and extending the framework in practice and identify avenues for explaining quality anomalies and prescribing solutions with business rules [57]. For example, process discovery in process mining could be used with BPI to retrieve the frequency counts of process model variants per day over time in BPM. Predicting these frequency counts over time, subtracting predicted frequency count from CTQs, converting these differences to z-scores, clustering the model variants based on these scores, and explaining predicted cluster memberships could serve as a starting point.

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